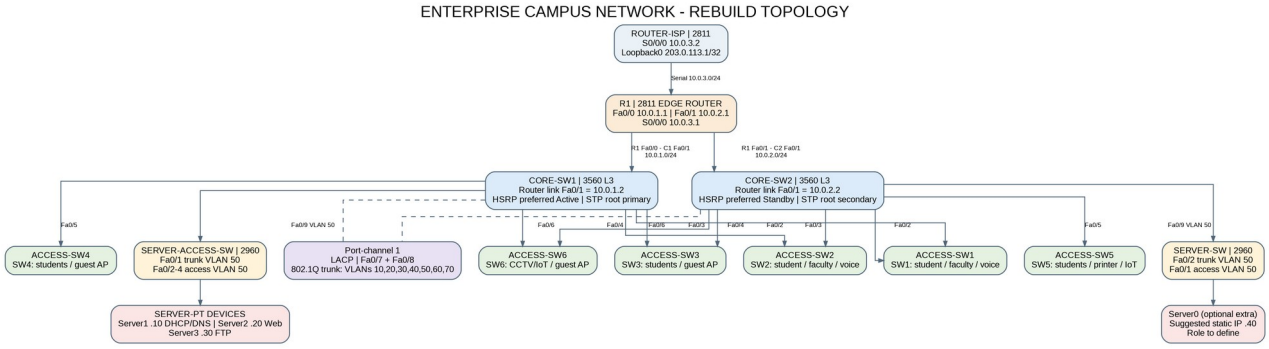
ENTERPRISE CAMPUS NETWORK

STEP-BY-STEP CONFIGURATION NOTES

Cisco Packet Tracer | Complete rebuild runbook



Purpose: This document is the practical rebuild recipe. Start with a blank Packet Tracer file and follow the stages in order. Each device has its own CLI block; server services use the Packet Tracer GUI.

Companion to: Enterprise Campus Network Complete Study Notes

How to use this runbook

1. Create a blank Packet Tracer file and place the devices listed in Stage 1.
2. Connect the cables using the exact local and remote interfaces in Stage 2. Do not guess port numbers.
3. Configure the core switches first, then routers, access switches, server switches and servers.
4. Paste only the block for the device currently open in the CLI. Never paste every block into one device.
5. After each device, save with copy running-config startup-config and run the verification commands listed.
6. Do not proceed past a failed verification. Fix the current stage first.

CLI prompt reminder: Commands such as ip route belong in global configuration mode: CORE-SW1(config)#. Interface commands belong under the correct interface prompt. If the CLI tries to translate a misspelled command or end as a hostname, press Ctrl+Shift+6 to cancel.

Prompt ending	Mode	Typical commands
>	User EXEC	Basic show commands; type enable to continue.
#	Privileged EXEC	show commands, copy running-config startup-config, configure terminal.
(config)#	Global configuration	hostname, vlan, ip route, ACL definitions, spanning-tree settings.
(config-if)#	Interface configuration	ip address, switchport mode, ip helper-address, no shutdown.
(config-ext-nacl)#	Extended ACL configuration	permit and deny statements.

Model-specific reminder: Cisco 2960 access switches use 802.1Q automatically. Do not enter switchport trunk encapsulation dot1q on a 2960. On a 3560 core switch, the encapsulation command is normally available; if Packet Tracer rejects it, omit that one line and configure switchport mode trunk.

Contents

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Important live port correction

Use the live rebuild mapping below: The current rebuild screenshots showed R1 Fa0/0 connected to CORE-SW1 Fa0/1, and R1 Fa0/1 connected to CORE-SW2 Fa0/1. An older note listed CORE-SW1 Fa0/0 as the router-facing port; that earlier port was rejected in the current Packet Tracer device. This runbook uses the port proven by CDP during the rebuild.

STAGE 1 - Create the Packet Tracer project

1.1 Device inventory

Device model	Quantity	Device names
Cisco 2811 Router	2	R1, ROUTER-ISP
Cisco 3560 Multilayer Switch	2	CORE-SW1, CORE-SW2
Cisco 2960 Switch	6	ACCESS-SW1 through ACCESS-SW6
Cisco 2960 Switch	2	SERVER-ACCESS-SW, SERVER-SW
Server-PT	4	Server1, Server2, Server3, Server0
PC / Laptop / Printer / Phone / AP / IoT	As required	Use the end-device port map in Stage 2

1. Open Cisco Packet Tracer and choose File > New.
2. Save As: Enterprise_Campus_Rebuild_v1.pkt. Keep your earlier Project.pkt as a backup.
3. Place two 2811 routers, two 3560 multilayer switches, six 2960 access switches and two 2960 server switches.
4. Place four Server-PT devices if you are keeping Server0 as well as Server1, Server2 and Server3.
5. Rename every device using the names in the inventory. The visible Packet Tracer label can be edited through the device Config/Display Name area; the IOS hostname is set by the CLI blocks.

1.2 Match the visible Packet Tracer icons to the hostnames

Visible workspace label in the rebuild	Hostname to configure
Router0 (lower router)	R1
Router0(1) (upper router)	ROUTER-ISP
Multilayer Switch0 (left core)	CORE-SW1
Multilayer Switch1 (right core)	CORE-SW2
Switch0 through Switch5 (six access switches)	ACCESS-SW1 through ACCESS-SW6, left-to-right as labeled SW1-SW6
Switch6 (left server switch)	SERVER-ACCESS-SW
Switch7 (right server switch)	SERVER-SW

Rename devices before cabling. The visible label helps identify the icon; the hostname set in IOS is what appears at the CLI prompt.

1.3 Router serial module

If the 2811 routers do not show Serial0/0/0, open each router > Physical, power it off, install a compatible WIC-2T serial module in an available slot, and power it back on. Confirm the serial interface exists before cabling.

Use the Packet Tracer serial DCE/DTE cable between R1 Serial0/0/0 and ROUTER-ISP Serial0/0/0. Only the DCE end needs a clock rate; identify it using show controllers serial 0/0/0 or the Packet Tracer cable/clock indication.

STAGE 2 - Exact cabling and interface map

2.1 Routers to core switches and ISP

Device A / port	Device B / port	Purpose
R1 Fa0/0	CORE-SW1 Fa0/1	Routed transit 10.0.1.0/24
R1 Fa0/1	CORE-SW2 Fa0/1	Routed transit 10.0.2.0/24
R1 S0/0/0	ROUTER-ISP S0/0/0	Serial transit 10.0.3.0/24

2.2 Core-to-access and server uplinks

CORE-SW1 local port	Connect to	CORE-SW2 local port	Connect to
Fa0/2	ACCESS-SW1 Fa0/1	Fa0/2	ACCESS-SW1 Fa0/2
Fa0/3	ACCESS-SW2 Fa0/1	Fa0/3	ACCESS-SW2 Fa0/2

CORE-SW1 local port	Connect to	CORE-SW2 local port	Connect to
Fa0/4	ACCESS-SW3 Fa0/1	Fa0/4	ACCESS-SW3 Fa0/2
Fa0/5	ACCESS-SW4 Fa0/1	Fa0/5	ACCESS-SW5 Fa0/1
Fa0/6	ACCESS-SW6 Fa0/1	Fa0/6	ACCESS-SW6 Fa0/2
Fa0/7	CORE-SW2 Fa0/7	Fa0/7	CORE-SW1 Fa0/7
Fa0/8	CORE-SW2 Fa0/8	Fa0/8	CORE-SW1 Fa0/8
Fa0/9	SERVER-ACCESS-SW Fa0/1	Fa0/9	SERVER-SW Fa0/2

The two core-to-core links Fa0/7 and Fa0/8 are later bundled into Port-channel 1. SW4 has one documented uplink to CORE-SW1. SW5 has one documented uplink to CORE-SW2.

2.3 End-device ports

Access switch	Port assignments
ACCESS-SW1	Fa0/3 PC0 VLAN30 | Fa0/4 Laptop0 VLAN30 | Fa0/5 Printer0 VLAN20 | Fa0/6 IP Phone0 data VLAN20 + voice VLAN70
ACCESS-SW2	Fa0/3 PC1 VLAN30 | Fa0/4 PC2 VLAN30 | Fa0/5 Printer1 VLAN20 | Fa0/6 IP Phone1 data VLAN20 + voice VLAN70
ACCESS-SW3	Fa0/3 PC3 VLAN30 | Fa0/4 PC4 VLAN30 | Fa0/5 Laptop1 VLAN30 | Fa0/6 Access Point0 VLAN40
ACCESS-SW4	Fa0/2-Fa0/4 student endpoints VLAN30 | Fa0/5 Access Point1 VLAN40
ACCESS-SW5	Fa0/2 PC7 VLAN30 | Fa0/3 PC8 VLAN30 | Fa0/4 Printer2 VLAN20 | Fa0/5 camera/IoT VLAN60
ACCESS-SW6	Fa0/3 camera/IoT VLAN60 | Fa0/4 Access Point VLAN40 | Fa0/5 camera/IoT VLAN60
SERVER-ACCESS-SW	Fa0/2 Server1 VLAN50 | Fa0/3 Server2 VLAN50 | Fa0/4 Server3 VLAN50
SERVER-SW	Fa0/1 Server0 VLAN50

Cable types: Use Copper Straight-Through or Packet Tracer automatic cable selection for ordinary Ethernet links. Use the serial DCE/DTE cable for the router serial connection. Green link indicators only show link state; they do not prove VLAN, routing or security configuration is correct.

STAGE 3 - VLAN and IP addressing plan

3.1 VLANs

VLAN ID	Name	Purpose	Network / mask	Virtual gateway
10	MANAGEMENT	Network-device management	10.10.10.0/24	10.10.10.1
20	FACULTY	Faculty / shared printers	10.10.20.0/24	10.10.20.1
30	STUDENTS	Student PCs and laptops	10.10.30.0/24	10.10.30.1
40	GUESTS	Guest wireless clients	10.10.40.0/24	10.10.40.1
50	SERVERS	Servers and applications	10.10.50.0/24	10.10.50.1
60	CCTV-IOT	Cameras and IoT devices	10.10.60.0/24	10.10.60.1
70	VOICE	IP phone voice traffic	10.10.70.0/24	10.10.70.1

3.2 Layer 3 transit networks

Link	Device/interface	IPv4 address	Subnet mask
R1 to CORE-SW1	R1 Fa0/0	10.0.1.1	255.255.255.0
R1 to CORE-SW1	CORE-SW1 Fa0/1	10.0.1.2	255.255.255.0
R1 to CORE-SW2	R1 Fa0/1	10.0.2.1	255.255.255.0
R1 to CORE-SW2	CORE-SW2 Fa0/1	10.0.2.2	255.255.255.0
R1 to ISP	R1 S0/0/0	10.0.3.1	255.255.255.0
R1 to ISP	ROUTER-ISP S0/0/0	10.0.3.2	255.255.255.0

3.3 HSRP physical and virtual addresses

Each core has its own SVI address. Hosts use the shared virtual address ending in .1 as their default gateway.

VLAN	CORE-SW1 SVI	CORE-SW2 SVI	Host default gateway
10	10.10.10.2	10.10.10.3	10.10.10.1
20	10.10.20.2	10.10.20.3	10.10.20.1
30	10.10.30.2	10.10.30.3	10.10.30.1
40	10.10.40.2	10.10.40.3	10.10.40.1
50	10.10.50.2	10.10.50.3	10.10.50.1
60	10.10.60.2	10.10.60.3	10.10.60.1
70	10.10.70.2	10.10.70.3	10.10.70.1

3.4 Server addressing

Server	IP address	Mask	Gateway	Role
Server1	10.10.50.10	/24	10.10.50.1	DHCP and DNS; NTP may be enabled if needed
Server2	10.10.50.20	/24	10.10.50.1	HTTP/HTTPS web server
Server3	10.10.50.30	/24	10.10.50.1	FTP file server
Server0	10.10.50.40*	/24	10.10.50.1	Optional fourth server; suggested address, role to define

Server0 note: 10.10.50.40 is a recommended unused static address for the additional Server0 shown in the topology. It was not present in the original server IP list. Use it only if Server0 remains in the rebuilt topology and confirm it is not already in use.

STAGE 4 - Configure CORE-SW1

CORE-SW1 is the primary multilayer core. It performs inter-VLAN routing, acts as the preferred HSRP Active gateway, is the STP root primary and connects to R1, access switches, EtherChannel and SERVER-ACCESS-SW.

4.1 Base, VLANs and STP root

```
enable
configure terminal
hostname CORE-SW1
ip routing

vlan 10
 name MANAGEMENT
vlan 20
 name FACULTY
vlan 30
 name STUDENTS
vlan 40
 name GUESTS
vlan 50
 name SERVERS
vlan 60
 name CCTV-IOT
vlan 70
 name VOICE

spanning-tree mode rapid-pvst
spanning-tree vlan 10,20,30,40,50,60,70 root primary
```

4.2 Router-facing routed interface

Important: Use Fa0/1 on CORE-SW1 for the R1 link in this rebuild. Do not use Fa0/0; your Packet Tracer device rejected that interface and CDP showed R1 Fa0/0 connected to CORE-SW1 Fa0/1.

```
interface fastethernet0/1
 no switchport
 ip address 10.0.1.2 255.255.255.0
 no shutdown
exit

ip route 0.0.0.0 0.0.0.0 10.0.1.1
```

4.3 Trunks to access switches

```
interface range fastethernet0/2 - 6
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,60,70
 no shutdown
exit
```

4.4 EtherChannel member ports and logical trunk

```
interface range fastethernet0/7 - 8
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,60,70
 channel-group 1 mode active
 no shutdown
exit

interface port-channel 1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,60,70
 no shutdown
exit
```

4.5 Server-switch trunk (VLAN 50 only)

```
interface fastethernet0/9
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 50
 no shutdown
exit
```

4.6 VLAN SVIs, HSRP and DHCP relay

```
interface vlan 10
 ip address 10.10.10.2 255.255.255.0
 standby 10 ip 10.10.10.1
 standby 10 priority 110
 standby 10 preempt
 ip helper-address 10.10.50.10
 no shutdown
exit

interface vlan 20
 ip address 10.10.20.2 255.255.255.0
 standby 20 ip 10.10.20.1
 standby 20 priority 110
 standby 20 preempt
 ip helper-address 10.10.50.10
 no shutdown
exit

interface vlan 30
 ip address 10.10.30.2 255.255.255.0
 standby 30 ip 10.10.30.1
 standby 30 priority 110
 standby 30 preempt
 ip helper-address 10.10.50.10
 no shutdown
exit

interface vlan 40
 ip address 10.10.40.2 255.255.255.0
 standby 40 ip 10.10.40.1
 standby 40 priority 110
 standby 40 preempt
 ip helper-address 10.10.50.10
 no shutdown
exit

interface vlan 50
 ip address 10.10.50.2 255.255.255.0
 standby 50 ip 10.10.50.1
 standby 50 priority 110
 standby 50 preempt
 no shutdown
exit

interface vlan 60
 ip address 10.10.60.2 255.255.255.0
 standby 60 ip 10.10.60.1
 standby 60 priority 110
 standby 60 preempt
 ip helper-address 10.10.50.10
 no shutdown
exit

interface vlan 70
 ip address 10.10.70.2 255.255.255.0
 standby 70 ip 10.10.70.1
 standby 70 priority 110
 standby 70 preempt
 ip helper-address 10.10.50.10
 no shutdown
exit
end
copy running-config startup-config
```

Why no helper on VLAN 50? Server1 at 10.10.50.10 is directly attached to VLAN 50. DHCP clients in other VLANs need the SVI helper address to relay their broadcasts to Server1.

4.7 Verify CORE-SW1

```
show ip interface brief
show ip route
show vlan brief
show interfaces trunk
show spanning-tree summary
show standby brief
show etherchannel summary
show cdp neighbors
```

STAGE 5 - Configure CORE-SW2

CORE-SW2 is the second multilayer core. It provides the redundant gateway, second routed link to R1, alternate access paths and the second side of EtherChannel.

5.1 Base, VLANs and STP secondary root

```
enable
configure terminal
hostname CORE-SW2
ip routing

vlan 10
 name MANAGEMENT
vlan 20
 name FACULTY
vlan 30
 name STUDENTS
vlan 40
 name GUESTS
vlan 50
 name SERVERS
vlan 60
 name CCTV-IOT
vlan 70
 name VOICE

spanning-tree mode rapid-pvst
spanning-tree vlan 10,20,30,40,50,60,70 root secondary
```

5.2 Router-facing routed interface

```
interface fastethernet0/1
 no switchport
 ip address 10.0.2.2 255.255.255.0
 no shutdown
exit

ip route 0.0.0.0 0.0.0.0 10.0.2.1
```

5.3 Trunks to access switches

```
interface range fastethernet0/2 - 6
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,60,70
 no shutdown
exit
```

5.4 EtherChannel matching configuration

```
interface range fastethernet0/7 - 8
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,60,70
 channel-group 1 mode active
 no shutdown
exit

interface port-channel 1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,60,70
 no shutdown
exit
```

5.5 Server-SW trunk

```
interface fastethernet0/9
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 50
 no shutdown
exit
```

5.6 VLAN SVIs, HSRP and DHCP relay

```
interface vlan 10
 ip address 10.10.10.3 255.255.255.0
 standby 10 ip 10.10.10.1
 standby 10 priority 100
 standby 10 preempt
 ip helper-address 10.10.50.10
 no shutdown
exit

interface vlan 20
 ip address 10.10.20.3 255.255.255.0
 standby 20 ip 10.10.20.1
 standby 20 priority 100
 standby 20 preempt
 ip helper-address 10.10.50.10
 no shutdown
exit

interface vlan 30
 ip address 10.10.30.3 255.255.255.0
 standby 30 ip 10.10.30.1
 standby 30 priority 100
 standby 30 preempt
 ip helper-address 10.10.50.10
 no shutdown
exit

interface vlan 40
 ip address 10.10.40.3 255.255.255.0
 standby 40 ip 10.10.40.1
 standby 40 priority 100
 standby 40 preempt
 ip helper-address 10.10.50.10
 no shutdown
exit

interface vlan 50
 ip address 10.10.50.3 255.255.255.0
 standby 50 ip 10.10.50.1
 standby 50 priority 100
 standby 50 preempt
 no shutdown
exit

interface vlan 60
 ip address 10.10.60.3 255.255.255.0
 standby 60 ip 10.10.60.1
 standby 60 priority 100
 standby 60 preempt
 ip helper-address 10.10.50.10
 no shutdown
exit

interface vlan 70
 ip address 10.10.70.3 255.255.255.0
 standby 70 ip 10.10.70.1
 standby 70 priority 100
 standby 70 preempt
 ip helper-address 10.10.50.10
 no shutdown
exit
end
copy running-config startup-config
```

5.7 Verify CORE-SW2

```
show ip interface brief
show ip route
show vlan brief
show interfaces trunk
show spanning-tree summary
show standby brief
show etherchannel summary
show cdp neighbors
```

STAGE 6 - Configure R1 and ROUTER-ISP

6.1 R1 edge-router configuration

```
enable
configure terminal
hostname R1

interface fastethernet0/0
 ip address 10.0.1.1 255.255.255.0
 ip nat inside
 no shutdown
exit

interface fastethernet0/1
 ip address 10.0.2.1 255.255.255.0
 ip nat inside
 no shutdown
exit

interface serial0/0/0
 ip address 10.0.3.1 255.255.255.0
 ip nat outside
 no shutdown
exit

ip route 10.10.0.0 255.255.0.0 10.0.1.2
ip route 10.10.0.0 255.255.0.0 10.0.2.2 10
ip route 0.0.0.0 0.0.0.0 10.0.3.2

access-list 1 permit 10.10.0.0 0.0.255.255
ip nat inside source list 1 interface serial0/0/0 overload

end
copy running-config startup-config
```

Meaning: R1 Fa0/0 and Fa0/1 face the campus and are marked ip nat inside. Serial0/0/0 faces the ISP and is marked ip nat outside. The first static route reaches campus through CORE-SW1; the second is a floating backup via CORE-SW2 with administrative distance 10. The default route sends unknown destinations toward the ISP. NAT overload lets many campus devices share R1 serial address 10.0.3.1.

6.2 ROUTER-ISP configuration

```
enable
configure terminal
hostname ROUTER-ISP

interface serial0/0/0
 ip address 10.0.3.2 255.255.255.0
 no shutdown
exit

interface loopback0
 ip address 203.0.113.1 255.255.255.255
 no shutdown
exit

ip route 10.10.0.0 255.255.0.0 10.0.3.1
end
copy running-config startup-config
```

Simulated Internet target: For this rebuild runbook, the test destination is 203.0.113.1 on ROUTER-ISP Loopback0. An older consolidated text note used 8.8.8.8 instead. Do not mix them: if you choose 8.8.8.8, change the loopback and ping target consistently.

6.3 Serial DCE clocking

If Serial0/0/0 remains up/down or down/down, identify which side is DCE. On that DCE interface only, enter:

```
configure terminal
interface serial0/0/0
 clock rate 64000
 no shutdown
end
```

6.4 Verify routers

```
show ip interface brief
show ip route
show controllers serial 0/0/0
show ip nat statistics
show ip nat translations
```

STAGE 7 - Configure ACCESS-SW1 through ACCESS-SW6

Common switch rule: Each access switch creates the VLANs it uses and enables Rapid PVST+. Trunks connect switches and carry multiple VLANs. Access ports connect a PC, laptop, printer, phone, camera or AP and belong to the assigned VLAN. Do not configure port security on trunk uplinks.

7.1 ACCESS-SW1

```
enable
configure terminal
hostname ACCESS-SW1
spanning-tree mode rapid-pvst
vlan 10
 name MANAGEMENT
vlan 20
 name FACULTY
vlan 30
 name STUDENTS
vlan 40
 name GUESTS
vlan 50
 name SERVERS
vlan 60
 name CCTV-IOT
vlan 70
 name VOICE

interface range fastethernet0/1 - 2
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,60,70
 no shutdown
exit

interface fastethernet0/3
 switchport mode access
 switchport access vlan 30
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 no shutdown
exit
interface fastethernet0/4
 switchport mode access
 switchport access vlan 30
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 no shutdown
exit
interface fastethernet0/5
 switchport mode access
 switchport access vlan 20
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 no shutdown
exit
interface fastethernet0/6
 switchport mode access
 switchport access vlan 20
 switchport voice vlan 70
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 no shutdown
exit
end
copy running-config startup-config
```

7.2 ACCESS-SW2

```
enable
configure terminal
hostname ACCESS-SW2
spanning-tree mode rapid-pvst
vlan 10
 name MANAGEMENT
vlan 20
 name FACULTY
vlan 30
 name STUDENTS
vlan 40
 name GUESTS
vlan 50
 name SERVERS
vlan 60
 name CCTV-IOT
vlan 70
 name VOICE

interface range fastethernet0/1 - 2
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,60,70
 no shutdown
exit
interface range fastethernet0/3 - 4
 switchport mode access
 switchport access vlan 30
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 no shutdown
exit
interface fastethernet0/5
 switchport mode access
 switchport access vlan 20
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 no shutdown
exit
interface fastethernet0/6
 switchport mode access
 switchport access vlan 20
 switchport voice vlan 70
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 no shutdown
exit
end
copy running-config startup-config
```

7.3 ACCESS-SW3

```
enable
configure terminal
hostname ACCESS-SW3
spanning-tree mode rapid-pvst
vlan 10
 name MANAGEMENT
vlan 20
 name FACULTY
vlan 30
 name STUDENTS
vlan 40
 name GUESTS
vlan 50
 name SERVERS
vlan 60
 name CCTV-IOT
vlan 70
 name VOICE

interface range fastethernet0/1 - 2
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,60,70
 no shutdown
exit
interface range fastethernet0/3 - 5
 switchport mode access
 switchport access vlan 30
 no shutdown
exit
interface fastethernet0/6
 switchport mode access
 switchport access vlan 40
 no shutdown
exit
end
copy running-config startup-config
```

7.4 ACCESS-SW4

```
enable
configure terminal
hostname ACCESS-SW4
spanning-tree mode rapid-pvst
vlan 10
 name MANAGEMENT
vlan 20
 name FACULTY
vlan 30
 name STUDENTS
vlan 40
 name GUESTS
vlan 50
 name SERVERS
vlan 60
 name CCTV-IOT
vlan 70
 name VOICE

interface fastethernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,60,70
 no shutdown
exit
interface range fastethernet0/2 - 4
 switchport mode access
 switchport access vlan 30
 no shutdown
exit
interface fastethernet0/5
 switchport mode access
 switchport access vlan 40
 no shutdown
exit
end
copy running-config startup-config
```

7.5 ACCESS-SW5

```
enable
configure terminal
hostname ACCESS-SW5
spanning-tree mode rapid-pvst
vlan 10
 name MANAGEMENT
vlan 20
 name FACULTY
vlan 30
 name STUDENTS
vlan 40
 name GUESTS
vlan 50
 name SERVERS
vlan 60
 name CCTV-IOT
vlan 70
 name VOICE

interface fastethernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,60,70
 no shutdown
exit
interface range fastethernet0/2 - 3
 switchport mode access
 switchport access vlan 30
 no shutdown
exit
interface fastethernet0/4
 switchport mode access
 switchport access vlan 20
 no shutdown
exit
interface fastethernet0/5
 switchport mode access
 switchport access vlan 60
 no shutdown
exit
end
copy running-config startup-config
```

7.6 ACCESS-SW6

```
enable
configure terminal
hostname ACCESS-SW6
spanning-tree mode rapid-pvst
vlan 10
 name MANAGEMENT
vlan 20
 name FACULTY
vlan 30
 name STUDENTS
vlan 40
 name GUESTS
vlan 50
 name SERVERS
vlan 60
 name CCTV-IOT
vlan 70
 name VOICE

interface range fastethernet0/1 - 2
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,60,70
 no shutdown
exit
interface fastethernet0/3
 switchport mode access
 switchport access vlan 60
 no shutdown
exit
interface fastethernet0/4
 switchport mode access
 switchport access vlan 40
 no shutdown
exit
interface fastethernet0/5
 switchport mode access
 switchport access vlan 60
 no shutdown
exit
end
copy running-config startup-config
```

7.7 Verify each access switch

```
show vlan brief
show interfaces trunk
show interfaces status
show spanning-tree summary
show cdp neighbors
```

For ACCESS-SW2, seeing Fa0/2 as Altn BLK in show spanning-tree vlan 30 can be normal. Fa0/1 is the Root FWD path and Fa0/2 is the alternate blocked path for that VLAN. STP blocks a redundant path to prevent loops; do not shut down or remove it just because it is blocking.

STAGE 8 - Configure the two dedicated server switches

8.1 SERVER-ACCESS-SW (Switch6 in the workspace)

This switch is connected to CORE-SW1 Fa0/9. Fa0/1 is the VLAN 50 trunk; Fa0/2-Fa0/4 are the three Server-PT access ports.

```
enable
configure terminal
hostname SERVER-ACCESS-SW
spanning-tree mode rapid-pvst
vlan 50
 name SERVERS
exit
interface fastethernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 50
 no shutdown
exit
interface range fastethernet0/2 - 4
 switchport mode access
 switchport access vlan 50
 no shutdown
exit
end
copy running-config startup-config
```

8.2 SERVER-SW (Switch7 in the workspace)

This switch is connected to CORE-SW2 Fa0/9. The actual CDP output showed CORE-SW2 Fa0/9 connected to SERVER-SW Fa0/2. Server0 connects to Fa0/1. Do not reverse these ports.

```
enable
configure terminal
hostname SERVER-SW
spanning-tree mode rapid-pvst
vlan 50
 name SERVERS
exit
interface fastethernet0/1
 switchport mode access
 switchport access vlan 50
 no shutdown
exit
interface fastethernet0/2
 switchport mode trunk
 switchport trunk allowed vlan 50
 no shutdown
exit
end
copy running-config startup-config
```

8.3 Verify server switches

```
show vlan brief
show interfaces trunk
show interfaces status
show cdp neighbors
```

STAGE 9 - Configure Server-PT devices and services

9.1 Server1 - DHCP and DNS server

1. Open Server1 > Desktop > IP Configuration.
2. Choose Static.
3. Enter IP Address 10.10.50.10, Subnet Mask 255.255.255.0, Default Gateway 10.10.50.1 and DNS Server 10.10.50.10.
4. Open Services > DHCP. Set Service to On.

5. Create the seven pools in the table below. Enter a pool name, gateway, DNS, start address, subnet mask and maximum number of users, then click Add for each pool.

Pool	Gateway	DNS	Start address	Mask	Max
MANAGEMENT	10.10.10.1	10.10.50.10	10.10.10.100	255.255.255.0	50
FACULTY	10.10.20.1	10.10.50.10	10.10.20.100	255.255.255.0	50
STUDENTS	10.10.30.1	10.10.50.10	10.10.30.100	255.255.255.0	50
GUESTS	10.10.40.1	10.10.50.10	10.10.40.100	255.255.255.0	50
SERVERS	10.10.50.1	10.10.50.10	10.10.50.100	255.255.255.0	50
CCTV-IOT	10.10.60.1	10.10.50.10	10.10.60.100	255.255.255.0	50
VOICE	10.10.70.1	10.10.50.10	10.10.70.100	255.255.255.0	50

Remove the accidental default pool: If a default pool such as serverPool remains with gateway 0.0.0.0, DNS 0.0.0.0 and a network address as the start IP, select it and remove it. Do not remove the seven correctly configured named pools.

9.2 DHCP relay on the core switches

DHCP Discover is a local broadcast. A Layer 3 boundary does not forward that broadcast by default. The SVI helper address relays the request to Server1 at 10.10.50.10. The same helper commands must be present on both core switches.

```
! Run on CORE-SW1 AND CORE-SW2
configure terminal
interface vlan 10
 ip helper-address 10.10.50.10
exit
interface vlan 20
 ip helper-address 10.10.50.10
exit
interface vlan 30
 ip helper-address 10.10.50.10
exit
interface vlan 40
 ip helper-address 10.10.50.10
exit
interface vlan 60
 ip helper-address 10.10.50.10
exit
interface vlan 70
 ip helper-address 10.10.50.10
exit
end
copy running-config startup-config
```

No helper is required on VLAN 50 because Server1 is already in VLAN 50. The server VLAN pool may exist, but the infrastructure servers keep their fixed static IPs.

9.3 Server2 - Web server

1. Open Server2 > Desktop > IP Configuration > Static.
2. Set 10.10.50.20 / 255.255.255.0, gateway 10.10.50.1, DNS 10.10.50.10.
3. Open Services > HTTP and set HTTP On. Keep HTTPS On if you want HTTPS testing.

9.4 Server3 - FTP file server

1. Open Server3 > Desktop > IP Configuration > Static.
2. Set 10.10.50.30 / 255.255.255.0, gateway 10.10.50.1, DNS 10.10.50.10.
3. Open Services > FTP and set FTP On.
4. Create a user such as networkuser and a lab password. Give only the required permissions, such as Read, Write, List, Delete and Rename.

9.5 Server0 - additional fourth server

If the extra Server0 remains connected to SERVER-SW, configure it as a static VLAN 50 host. Suggested address: 10.10.50.40 / 255.255.255.0, gateway 10.10.50.1, DNS 10.10.50.10. Its application role is not specified in the rebuild notes; configure a service only if you need one.

9.6 DNS records on Server1

1. Open Server1 > Services > DNS and set DNS Service On.
2. Add A record: www.campus.local -> 10.10.50.20.
3. Add A record: files.campus.local -> 10.10.50.30.
4. Click Add after entering each record.

9.7 Client IP settings

For PCs/laptops that should use DHCP: open the device > Desktop > IP Configuration > DHCP. Expected example for PC0 or PC7 in VLAN 30: address in 10.10.30.100-149 (depending on prior leases), mask 255.255.255.0, gateway 10.10.30.1, DNS 10.10.50.10. Printers can use DHCP or a documented static address; avoid duplicate addresses.

NTP note: Server1 was originally planned to provide NTP as well, but the rebuild conversation did not confirm a completed NTP service configuration or client synchronization. Enable Services > NTP only if you intend to use it, and then configure/verify clients separately.

STAGE 10 - Configure ACLs, SSH and Port Security

10.1 Student ACL - block VLAN30 to VLAN10

Create and apply this ACL on BOTH CORE-SW1 and CORE-SW2. It blocks Student subnet 10.10.30.0/24 from Management subnet 10.10.10.0/24; it permits other IPv4 traffic through this ACL.

```
enable
configure terminal
ip access-list extended STUDENT-RESTRICTION
 deny ip 10.10.30.0 0.0.0.255 10.10.10.0 0.0.0.255
 permit ip any any
exit
interface vlan 30
 ip access-group STUDENT-RESTRICTION in
exit
end
copy running-config startup-config
```

10.2 Guest ACL - block guest to internal networks, keep DHCP and DNS

Apply on BOTH core switches. The DHCP exception allows a guest client DHCP request (UDP source port 68 to destination port 67). The DNS exceptions permit guests to query Server1 at 10.10.50.10. Other guest traffic to the internal 10.10.0.0/16 space is denied.

```
enable
configure terminal
ip access-list extended GUEST-RESTRICTION
 permit udp any eq 68 any eq 67
 permit udp 10.10.40.0 0.0.0.255 host 10.10.50.10 eq 53
 permit tcp 10.10.40.0 0.0.0.255 host 10.10.50.10 eq 53
 deny ip 10.10.40.0 0.0.0.255 10.10.0.0 0.0.255.255
 permit ip any any
exit
interface vlan 40
 ip access-group GUEST-RESTRICTION in
exit
end
copy running-config startup-config
```

ACL order matters: Cisco ACL entries are processed top-to-bottom and stop at the first match. Put DHCP/DNS permits before the broad internal-network deny. There is also an implicit deny at the end of every ACL; the final permit ip any any is what allows other IPv4 destinations.

10.3 SSH - enable remote management on routers and core switches

Repeat this block separately on CORE-SW1, CORE-SW2, R1 and ROUTER-ISP. Change the hostname line to the device name. Use your own lab password if preferred.

```
enable
configure terminal
ip domain-name campus.local
username netadmin secret NetLab@123
crypto key generate rsa
! When prompted, enter 1024 for the modulus if supported
ip ssh version 2
line vty 0 4
 login local
 transport input ssh
exit
end
copy running-config startup-config
```

Password safety: NetLab@123 is the example lab password used during this rebuild. Do not reuse it for production or a real network. Replace it before sharing the .pkt file or configuration backups.

10.4 Port Security - ACCESS-SW1

```
enable
configure terminal
interface fastethernet0/3
 switchport mode access
 switchport access vlan 30
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 no shutdown
exit
interface fastethernet0/4
 switchport mode access
 switchport access vlan 30
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 no shutdown
exit
interface fastethernet0/5
 switchport mode access
 switchport access vlan 20
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 no shutdown
exit
interface fastethernet0/6
 switchport mode access
 switchport access vlan 20
 switchport voice vlan 70
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 no shutdown
exit
end
copy running-config startup-config
```

10.5 Port Security - ACCESS-SW2

```
enable
configure terminal
interface fastethernet0/3
 switchport mode access
 switchport access vlan 30
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 no shutdown
exit
interface fastethernet0/4
 switchport mode access
 switchport access vlan 30
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 no shutdown
exit
interface fastethernet0/5
 switchport mode access
 switchport access vlan 20
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 no shutdown
exit
interface fastethernet0/6
 switchport mode access
 switchport access vlan 20
 switchport voice vlan 70
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 no shutdown
exit
end
copy running-config startup-config
```

Port Security limits the source MAC addresses learned on a protected access port. Sticky mode learns the connected MAC and adds it to the running configuration. Maximum 2 on the phone port allows the IP phone plus a downstream PC. Do not apply port security to Fa0/1 or Fa0/2 trunk uplinks.

STAGE 11 - Final verification and test plan

11.1 Switch verification

```
show vlan brief
show interfaces trunk
show interfaces status
show cdp neighbors
show spanning-tree summary
show spanning-tree vlan 30
```

11.2 Core verification

```
show ip interface brief
show ip route
show standby brief
show etherchannel summary
show access-lists
show ip interface vlan 30
show ip interface vlan 40
```

11.3 Router verification

```
! On R1
show ip interface brief
show ip route
show ip nat statistics
show ip nat translations
show access-lists 1

! On ROUTER-ISP
show ip interface brief
show ip route
```

11.4 Required expected results

Test	Command / action	Expected result
Router-to-core	R1 ping 10.0.1.2 and 10.0.2.2	Both directly connected core links reply
Serial link	R1 ping 10.0.3.2	ISP serial peer replies
Student DHCP	PC0 > Desktop > IP Configuration > DHCP	10.10.30.x, mask /24, gateway .30.1, DNS .50.10
Second-core DHCP	PC7 > Desktop > IP Configuration > DHCP	Receives an address in VLAN30 through the second core path
Gateway	PC0 ping 10.10.30.1	Reply from HSRP virtual gateway
Server reachability	PC0 ping 10.10.50.10 / .20 / .30	Replies if routes and ACL allow the traffic
DNS	ping www.campus.local	Resolves to 10.10.50.20
Web	Browser: http://www.campus.local	Server2 web page appears
FTP	ftp files.campus.local	Server3 accepts configured user login
Student ACL	PC0 ping 10.10.10.2	Blocked by STUDENT-RESTRICTION
Student allowed destination	PC0 ping 10.10.50.10	Permitted by student ACL if server path is healthy
Guest ACL	Guest client ping 10.10.10.2 and 10.10.50.20	Blocked by GUEST-RESTRICTION
Guest DNS	Guest DNS lookup for www.campus.local	DNS to Server1 is allowed by ACL
NAT/PAT	PC0 ping 203.0.113.1; then R1 show ip nat translations	Ping succeeds and a translation is created
SSH	PC0 command prompt: ssh -l netadmin 10.0.1.1	Login prompt and remote R1 CLI
Port Security	show port-security address on SW1/SW2	Sticky MAC addresses appear after traffic is sent

11.5 HSRP failover test

1. On CORE-SW1, verify show standby brief. VLAN30 should show CORE-SW1 Active and CORE-SW2 Standby.
2. Temporarily shut only interface vlan 30 on CORE-SW1: configure terminal > interface vlan 30 > shutdown.
3. On CORE-SW2, run show standby brief. VLAN30 should become Active on CORE-SW2.
4. Restore CORE-SW1: configure terminal > interface vlan 30 > no shutdown.
5. Run show standby brief on both cores. With higher priority and preempt configured, CORE-SW1 should return to Active.

Do not shut down the whole switch for this basic test: The controlled test shuts only the VLAN 30 SVI. Restore it immediately after confirming takeover. This verifies one VLAN gateway group, not every VLAN group.

11.6 EtherChannel success criteria

```
show etherchannel summary
```

Expected line on BOTH cores: Port-channel 1 should show Po1(SU), protocol LACP, and Fa0/7(P) Fa0/8(P). S means Layer 2, U means in use and P means the member port is bundled. If the output says Po1(SD) or member ports (s), do not declare EtherChannel operational; compare trunk VLAN list, mode, speed/duplex and channel-group on both sides.

11.7 STP interpretation

CORE-SW1 is configured root primary and CORE-SW2 root secondary. Access switches use Rapid PVST+ but are not configured as roots. In a redundant design, STP may show alternate blocking ports. A line such as ACCESS-SW2 Fa0/2 Altn BLK for VLAN 30 can be normal when Fa0/1 is the root forwarding path. Blocking prevents a Layer 2 loop.

STAGE 12 - Troubleshooting guide

Symptom	First checks	Likely correction
Interface command invalid	show ip interface brief; show cdp neighbors	Use the actual interface shown by CDP. In this rebuild CORE-SW1 router link is Fa0/1, not Fa0/0.
Port-channel suspended	show etherchannel summary; show interfaces trunk; show run	Both cores must match channel-group 1 mode active, trunk mode, allowed VLAN list and physical port mapping.
SVI down/down	show ip interface brief; show vlan brief; show interfaces trunk	VLAN must exist, be carried on at least one active Layer 2 port, and interface vlan must not be shut down.
HSRP standby unknown	show standby brief; show spanning-tree vlan N	Check matching group, virtual IP, SVI addresses, VLAN path and trunk allowed lists.
DHCP fails	Server DHCP On; pool values; ip helper-address on both cores	Confirm correct pool gateway, start address, DNS, mask and helper IP 10.10.50.10.
DNS name fails but IP ping works	Server1 Services > DNS; client DNS field	Turn DNS On, confirm A record, confirm DHCP scope advertises 10.10.50.10 as DNS.
Web/FTP fails	Server service On; static IP; client DNS; ACL	Check HTTP/FTP service, server VLAN path, DNS record and ACL exceptions.
NAT table empty	ping external target, show ip nat statistics	Generate traffic; confirm R1 inside/outside labels, ACL 1, default route and ISP return route.
Serial up/down	show controllers serial 0/0/0	Set clock rate on DCE side only; verify serial cable and both IPs/subnet.
Port-security violation	show port-security interface fa0/x	Check unexpected MAC, maximum count, phone plus PC MACs; recover a disabled port only after correcting the cause.

APPENDIX A - SSH test commands

From a client PC command prompt, use the configured local username and lab password. Example:

```
ssh -l netadmin 10.0.1.1   ! R1
ssh -l netadmin 10.0.3.2   ! ROUTER-ISP
ssh -l netadmin 10.10.10.2 ! CORE-SW1 management SVI
ssh -l netadmin 10.10.10.3 ! CORE-SW2 management SVI
```

A student PC is blocked from accessing VLAN10 by the Student ACL, so a student endpoint is not the right test source for SSH to the core management addresses. Use a management VLAN host or a permitted administrator test host.

APPENDIX B - Access-switch management IPs (optional)

The main rebuild did not require individual management IPs on all access switches. If you want to manage the six access switches by SSH later, assign a unique VLAN10 SVI address and default gateway on each. VLAN10 must exist and be allowed on that switch uplink.

Switch	Suggested VLAN10 management IP	Default gateway
ACCESS-SW1	10.10.10.11	10.10.10.1
ACCESS-SW2	10.10.10.12	10.10.10.1
ACCESS-SW3	10.10.10.13	10.10.10.1
ACCESS-SW4	10.10.10.14	10.10.10.1
ACCESS-SW5	10.10.10.15	10.10.10.1
ACCESS-SW6	10.10.10.16	10.10.10.1

```
! Example for ACCESS-SW1 only - change the address for each switch
configure terminal
interface vlan 10
 ip address 10.10.10.11 255.255.255.0
 no shutdown
exit
ip default-gateway 10.10.10.1
end
copy running-config startup-config
```

Do not apply this VLAN10 management SVI to SERVER-ACCESS-SW or SERVER-SW without changing their uplinks: Their current uplinks are restricted to VLAN50 only, so VLAN10 would not reach the core over those links.

APPENDIX C - Useful command dictionary

Command	What it tells you
show vlan brief	VLAN database and access-port membership
show interfaces trunk	Which interfaces trunk and which VLANs are allowed/active
show cdp neighbors	Direct Cisco neighbors and local/remote port mapping
show spanning-tree vlan 30	Root bridge, root port, designated and alternate/blocking ports for VLAN30
show etherchannel summary	Port-channel state, protocol and bundled member ports
show standby brief	HSRP group, virtual IP, Active and Standby roles
show ip interface brief	Interface IP addresses and line/protocol status
show ip route	Connected, static, default and learned routes
show access-lists	ACL rules and match counters
show ip nat translations	Current NAT/PAT translation entries
show ip ssh	SSH version and server status
show port-security address	MAC addresses learned by Port Security

APPENDIX D - Final rebuild checklist

☐ New .pkt file saved under a new name; original project kept as backup.

☐ All devices placed, renamed, and cabled to the correct interface numbers.

☐ R1, ROUTER-ISP, CORE-SW1 and CORE-SW2 IP addressing verified.

☐ VLANs 10, 20, 30, 40, 50, 60 and 70 exist on the required switches.

☐ All access trunks and the VLAN50 server trunks verified.

☐ Port-channel 1 shows Po1(SU), Fa0/7(P), Fa0/8(P) on both cores.

☐ Rapid PVST+ is enabled; CORE-SW1 root primary and CORE-SW2 root secondary.

☐ HSRP Active/Standby verified across VLANs 10-70; failover and restore tested.

☐ R1 and ISP routes verified; serial link is up/up.

☐ Server1 static IP, DHCP pools and DHCP relay on both cores configured.

☐ Server2 HTTP and Server3 FTP configured; DNS A records added to Server1.

☐ PC0 and PC7 receive valid DHCP addresses from VLAN30.

☐ Student and Guest ACLs applied on both core switches and tested.

☐ SSH works on the intended infrastructure devices.

☐ Port Security verified on ACCESS-SW1 and ACCESS-SW2.

☐ NAT/PAT ping to the simulated ISP loopback succeeds and translation appears.

☐ All running configurations saved with copy running-config startup-config.

☐ Final .pkt, topology screenshot, show-command outputs and README backed up.

Important scope and rebuild notes

- This runbook contains the device configuration and Packet Tracer GUI steps used for the rebuild. The companion Complete Study Notes explain the concepts in more depth.
- The exact current rebuild mapping is used where a screenshot/CDP output corrected older notes. Specifically, CORE-SW1 router uplink uses Fa0/1.
- The ISP loopback in this runbook is 203.0.113.1/32 because that was the test target used during the latest rebuild conversation. Do not mix it with the older 8.8.8.8 example.
- The extra Server0 address 10.10.50.40 is suggested, not a previously confirmed address. Server0 role was not fully specified.
- NTP was originally listed as an intended Server1 service but was not confirmed as enabled or tested. Phone registration/calling also needs Call Manager/CME/TFTP configuration beyond DHCP.
- Access-switch management IPs in Appendix B are optional suggested addresses, not required for the core VLAN, DHCP, routing or end-user data path.
- A configuration command being entered does not by itself prove the feature is working. Use the verify commands and expected outputs in Stage 11.